

- ৯। (ক) ল্যাঞ্জেভিন তত্ত্ব থেকে উপচুম্বকতা বিষয়ক কুরির সূত্র ব্যুৎপন্ন করো।
- ✓✓ (খ) (অ) ডোমেন বা অঞ্চল কী?
- ✓✓ (আ) তিরশ্চৌম্বক পদার্থের মধ্যে অঞ্চলের অস্তিত্ব ভৌতভাবে ব্যাখ্যা করো।
- ✓✓ (ই) উপরের প্রশ্নের উত্তরের স্বপক্ষে সরাসরি পরীক্ষামূলক প্রমাণ দাও।
- ✓✓ (ঈ) অঞ্চল ধারণার ব্যবহারিক প্রয়োগের উদাহরণ দাও।
- ✓✓ (গ) ঘরের উষ্ণতায় Fe_2O_3 -র উপচৌম্বকগ্রাহীতা 1.4×10^{-3} । প্রতিটি পরমাণুর দ্বিমেরু ভ্রামকের মান যদি বোর ম্যাগনেটনের ক্রমে হয়, তবে পদার্থটির প্রতি একক আয়তনে পরমাণুর সংখ্যা গণনা করো। দেওয়া আছে, এক বোর ম্যাগনেটন $= 9.27 \times 10^{-24} \text{ Am}^2$ ।

$$8 + (1 + 2 + 1 + 1) + 3$$

[English Version]

The figures in the margin indicate full marks.

Answer **question no. 1** which is compulsory and **any five** questions, taking at least **one** question from each **Group**.

1. Answer **any five** questions :

3×5

- Spacecraft A is moving at $0.90c$ with respect to the earth. If spacecraft B is to pass A at a relative speed of $0.50c$ in same direction, what speed must B have with respect to the earth?
- Find the wavelength and frequency of a 100 MeV photon.
- Find the de Broglie wavelength of a 46 g golf ball with a velocity of 30 m/s .
- An eigenfunction of the operator $\frac{d^2}{dx^2}$ is $\psi = e^{2x}$. Find the corresponding eigenvalue.
- Evaluate the commutator $[L\hat{x}, p\hat{x}]$.
- Give a comparative study of dia, para and ferromagnetism.
- Explain the usefulness of B-H curves.
- The first order reflection angle is 18° from (100) plane of a cubic crystal using X-rays of wavelength $1.54 \text{ \AA}$. Determine the distance between the (100) planes and the (111) planes of crystal.

Group - A

- Prove that if $\frac{v}{c} \ll 1$, the kinetic energy K of a moving particle will always be much less than its rest energy m_0c^2 .
 - At what speed v will the Galilean and Lorentz expressions for length differ by 1%?

1. Answer *any five* questions :

3×5

- (a) Spacecraft A is moving at $0.90c$ with respect to the earth. If spacecraft B is to pass A at a relative speed of $0.50c$ in same direction, what speed must B have with respect to the earth?
- (b) Find the wavelength and frequency of a 100 MeV photon.
- (c) Find the de Broglie wavelength of a 46 g golf ball with a velocity of 30 m/s.
- (d) An eigenfunction of the operator $\frac{d^2}{dx^2}$ is $\psi = e^{2x}$. Find the corresponding eigenvalue.
- (e) Evaluate the commutator $[L\hat{x}, p\hat{x}]$.
- (f) Give a comparative study of dia, para and ferromagnetism.
- (g) Explain the usefulness of B-H curves.
- (h) The first order reflection angle is 18° from (100) plane of a cubic crystal using X-rays of wavelength $1.54 \text{ \AA}$. Determine the distance between the (100) planes and the (111) planes of crystal.

Group - A

2. (a) Prove that if $\frac{v}{c} \ll 1$, the kinetic energy K of a moving particle will always be much less than its rest energy m_0c^2 .
- (b) At what speed v will the Galilean and Lorentz expressions for length differ by 1%?

(c) The average lifetime of μ -mesons at rest is 2.3×10^{-6} s. A laboratory measurement on μ -mesons yields an average lifetime of 6.9×10^{-6} s. What is the speed of the mesons in the laboratory?

(d) Plot the functions $\sqrt{1-s^2}$ and $\gamma = \frac{1}{\sqrt{1-s^2}}$ as a function of s . 2+2+4+(2+2)

3. (a) State Planck's law of blackbody radiation.

(b) Draw the curve of stopping potential with the frequency of incident light.

(c) Explain the physical significance of Compton effect. Derive the formula of Compton shift

$$\Delta\lambda = \frac{h}{m_0c}(1 - \cos\phi), \text{ where } \phi \text{ is scattering angle.} \quad 2\frac{1}{2}+2\frac{1}{2}+(2+5)$$

4. (a) Show that the ratio of the de Broglie wavelength to the Compton wavelength of a particle is

$$\sqrt{\left(\frac{C}{v}\right)^2 - 1}.$$

(b) An electron in an atom is excited to next higher energy state. The electron stays in that excited state for 10^{-8} s and then goes to its previous state; in this process, the excess energy is emitted in the form of a Photon. Find the inherent uncertainty in the frequency of the Photon.

(c) A particle moving in a straightline is described by

$$\psi(x) = \frac{(x+ix)}{(1+ix^2)}.$$

Obtain $\psi^*(x)$.

(d) Normalise the following wave function in one-dimension :

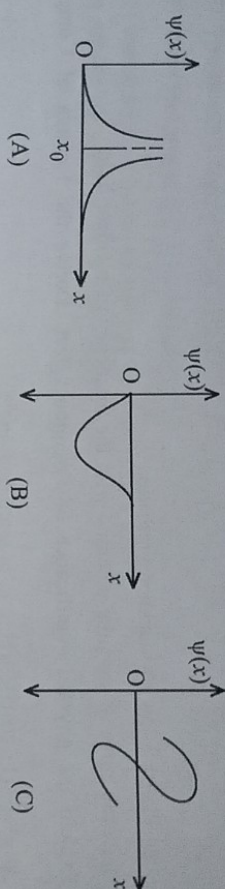
$$\begin{aligned} \psi(x) &= Ae^{-\alpha x}, \text{ for } x > 0 \\ &= Ae^{+\alpha x}, \text{ for } x < 0, \end{aligned}$$

where α is a positive constant.

3+3+2+4

Group - B

5. (a) Which of the wave functions is shown in Figures do not have physical significance in the interval shown? Why?



- (b) Calculate the following probability density for the wave function :

$$\Psi(x,t) = Ae^{-\left(\sqrt{am}/2h\right)x^2 - \left(\frac{i}{2}\right)\left(\frac{c}{m}\right)t}$$

- (c) (i) Apply Schrödinger equation to determine the energy eigenvalue, E for the wave function

$$\Psi(x,t) = A \cos \frac{\pi x}{a} e^{-iEt/\hbar}, \quad \left(-\frac{a}{2} < x < +\frac{a}{2}\right)$$

$$= 0, \quad \left(x \leq -\frac{a}{2} \text{ or } x \geq +\frac{a}{2}\right),$$

where A is an arbitrary constant, E is the total energy of the particle.

- (ii) Plot the ground state wave function, for the above problem.

3+3+(4+2)

6. (a) ~~Derive~~ time independent one-dimensional Schrödinger wave equation.

- (b) State the physical interpretation of the wave function.

- (c) ~~Normalize~~ the following wave function :

$$\psi_n(x) = A \sin \frac{n\pi x}{L} \text{ for particle confined in a length } 0 < x < L.$$

- (d) Find the eigenenergy value (E_n) for n -th state.

4+2+3+3

7. (a) (i) Find the Miller indices of a plane (in a crystal) having intercepts of $6a$, $4b$, $2c$ on the x , y and z -axes respectively.

- (ii) Determine also the intercepts of two other planes one on each side of this plane and having these indices.

- (b) Determine the interplanar spacing of plane (hkl) when the lattice is a cube of edge length a .

- (c) The wavelength of the K_{α} line of Ag is $0.563 \text{ \AA}$. The radiation from a Ag target is analysed with a Bragg Spectrometer using a Calcite crystal (a simple cube of lattice constant $3.02945 \text{ \AA}$).
- Determine the angle of reflection for the first order.
 - What is the highest order for which this one may be observed? (2+3)+3+(2+2)
8. (a) Explain with the help of a diagram the process of bond formation between Magnesium and Oxygen. Name the bond.
- (b) Arrange the following bonds in ascending order of their strength and write an example of each :
 (i) Ionic bond (ii) van der Waals bond (iii) Covalent bond.
- (c) What is bandgap? Compare the bandgap values of conductors, semiconductors and insulators.
- (d) What is the role of dopant concentration on the Fermi energy of n - and p -type semiconductors? (2+1)+3+(1+3)+2
9. (a) Derive Curie's law of paramagnetism from Langevin's theory.
- (b) (i) What are domains? 2+2+1+1+1+1
 (ii) Explain the existence of domains in a ferromagnetic material.
 (iii) Give a direct experimental evidence in support of your answer.
 (iv) Also enumerate a practical application of the concept of domains.
- (c) The paramagnetic susceptibility of Fe_2O_3 at room temperature is 1.4×10^{-3} . If the dipole moment of each atom is of the order of a Bohr magneton, calculate the number of atoms per unit volume of the material. Given, $1 \text{ Bohr magneton} = 9.27 \times 10^{-24} \text{ Am}^2$. 4+(1+2+1+1)+3